

# Anik Pramanik

Newark, NJ 07104 • [anikpramanikcse@gmail.com](mailto:anikpramanikcse@gmail.com) • +1 551 292 0887  
Website • [Google Scholar](#) • [LinkedIn](#) • [GitHub](#)

## Research Interests

---

LLM Agents (Text-to-SQL Systems); LLM Security; Federated Learning

## Education

---

**New Jersey Institute of Technology (NJIT)**, Newark, NJ  
Ph.D. in Computer Science, Advisor: [Shantanu Sharma](#)

Sep 2021 – Dec 2026 (Expected)

**Khulna University of Engineering & Technology**, Bangladesh  
B.Sc. in Computer Science and Engineering

Mar 2016 – Mar 2020

## Selected Publications

---

- **A. Pramanik**, M. Kantarcioglu, V. Oria, S. Sharma, *DexterSQL: Deep Schema Exploration and Rule-based Correction for Text-to-SQL Generation*, Preprint, under review, 2026. ([pdf](#))
- **A. Pramanik**, M. Kantarcioglu, V. Oria, S. Sharma, *FedDAG: Clustered Federated Learning via Global Data and Gradient Integration for Heterogeneous Environments*, **ICLR** 2026. ([pdf](#))
- **A. Pramanik**, M. Kumar, N. Panwar, L. V. Potts, J. Shah, S. Sharma, *Spark: Smart Building Fire Prediction and Risk Analysis*, **IEEE NCA** 2025. ([pdf](#))
- **A. Pramanik**, P. Xu, Y. Xu, *Equity Promotion in Public Transportation*, **AAAI** 2023. ([pdf](#))
- **A. Pramanik**, A. Sarker, *Parkinson's Disease Detection from Voice and Speech Data Using Machine Learning*, **IJCACI** 2021. ([pdf](#))

## Research Experience

---

**Doctoral Research Assistant, New Jersey Institute of Technology**

Sep 2021 – Present

- **LLM Security/Safety**: Developing a secure agentic Text-to-SQL system that defends against prompt-injection attacks and enforces dynamic, fine-grained access control, limiting query generation and execution to each user's authorized data.
- **Agentic LLM Systems**: Developed a multi-stage agentic Text-to-SQL system in Python combining prompt engineering, retrieval, LLM evaluation, and tool calling; improved execution accuracy by  $\approx 3\%$  over SOTA approaches.
- **LLM Fine-tuning & Alignment**: Investigated fine-tuning of 10B+ models for Text-to-SQL using a LoRA/SFT warm start followed by policy-gradient reinforcement learning with execution-based rewards, guided by error analysis.
- **Distributed ML & Federated Learning**: Proposed FedDAG, an FL framework in PyTorch with a dual-encoder architecture and graph-guided feature sharing over non-IID data, improving by  $\approx 4\%$  over SOTA on vision benchmarks.
- **Large-Scale LLM Experimentation**: Evaluated agentic Text-to-SQL methods with open-weight LLMs up to 120B on Slurm-managed multi-A100 systems across model sizes and inference configurations.
- **Optimization & Resource Allocation**: Formulated equity-aware public-transit resource allocation as an integer program and designed an LP-relaxation randomized-rounding algorithm with a provable  $1 - 1/e$  approximation guarantee.

## Teaching Experience

---

**Graduate Teaching Assistant, New Jersey Institute of Technology**

Sep 2021 – Present

- Supported 25+ course sections in AI, ML, DBMS, and DSA through laboratory instruction, office hours, grading, and one-on-one mentoring, helping students connect core concepts with practical implementations.

## Honors & Awards

---

- Nominated for Full Membership, **Sigma Xi, The Scientific Research Honor Society**, 2026.
- **Graduate Student Assistantship**, NJIT, Fall 2021 – Present. Full tuition support and stipend for Ph.D. study.
- Achieved **Codeforces Expert** rating, max rating 1621, & solved 500+ problems: *CF handle*: [anik\\_1507103](#)
- **Vocational Scholarship**, awarded for the first two years of undergraduate study at KUET.

## Professional Service

---

**Reviewer**: NeurIPS 2026, AAAI 2026; ACM CODASPY 2025; IEEE CIC 2025

## Technical Skills

---

**Languages:** Python, SQL, C/C++, Java, Bash, JavaScript

**LLM & Agents:** Retrieval-Augmented Generation (RAG), prompt engineering, MCP, tool calling, LLM evaluations

**LLM Fine-tuning:** PEFT/LoRA, SFT, DPO, ORPO, RLHF, PPO, GRPO, reward modeling, Hugging Face

**Training & Inference Optimization:** PyTorch, TensorFlow, Keras, Scikit-learn, CUDA, FSDP, DeepSpeed, vLLM, Slurm/HPC

**Data & Infrastructure:** Pandas, NumPy, PostgreSQL, MySQL, Oracle, AWS, Docker, Git, Linux, REST APIs

## Selected Projects

---

### Agentic Text-to-SQL

<https://github.com/anikpramanikcse/DexterSQL>

*Python, SQL, FAISS, SQLGlue, spaCy, Sentence Transformers, API*

- Built a prompting-based Text-to-SQL system with deep targeted schema exploration and dedicated schema linking; applied multi-path SQL generation, SQL selection, and rule-based correction to improve execution accuracy.
- Analyzed ambiguous columns and individual/joint data distributions to resolve schema ambiguity; mined correction rules from training data and used query dependency trees to ground SQL generation.

### Federated Learning on Heterogeneous Data

<https://github.com/anikpramanikcse/FedDAG>

*PyTorch, Torchvision, NumPy, SciPy, Scikit-learn, Matplotlib*

- Improved clustered FL on heterogeneous non-IID data by fusing class-wise data and sparse-gradient similarities; used a federation-aware predictor metric to evaluate candidate clusterings.
- Designed a dual-encoder clustered-FL architecture for cross-cluster gradient sharing, importing missing low-level representations while preserving cluster specialization and data privacy.

## References

---

### Shantanu Sharma

Assistant Professor, NJIT  
[shantanu.sharma@njit.edu](mailto:shantanu.sharma@njit.edu)

### Murat Kantarcioglu

Professor, Virginia Tech  
[muratk@vt.edu](mailto:muratk@vt.edu)

### Vincent Oria

Professor, NJIT  
[vincent.oria@njit.edu](mailto:vincent.oria@njit.edu)